Chapter 1
Statistical Estimation Theory

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1.1 Estimation Theory

Estimation theory plays a central role in many data science applications by providing foundational methods for inferring unknown parameters from observed data. Estimation techniques are used to design algorithms that learn from data and make predictions about unseen samples. In signal processing, it helps recover signals from noisy measurements, while in statistical inference, it forms the basis for hypothesis testing and parameter estimation.

1.1.1 The estimation problem

In its most general form, the estimation problem consists of designing a function, known as an *estimator*, that maps an input vector of observations to an estimate of a target variable. This target may represent an unknown physical quantity, a model parameter, or a latent variable in a probabilistic model.

Mathematically, we denote the observed data as a random vector $\mathbf{X}$, drawn from some observation space $\mathscr{X}$, and the target variable to be estimated as S. We will generally assume that the target variable is a scalar, although we will analyze some multidimensional scenarios.[1]

The goal of estimation is to construct a function $f : \mathscr{X} \to \mathbb{R}$, called the *estimator*, that produces an estimate $\hat{S} = f(\mathbf{X})$ of the unknown S, named the *estimation* or *prediction*. The estimator is a deterministic function, meaning that for a given input $\mathbf{x}$, it will consistently produce the same output. However, when the input $\mathbf{X}$ is random vector, the prediction $\hat{S}$ is also a random variable.

Figure 1.1 provides a schematic representation of the estimation problem

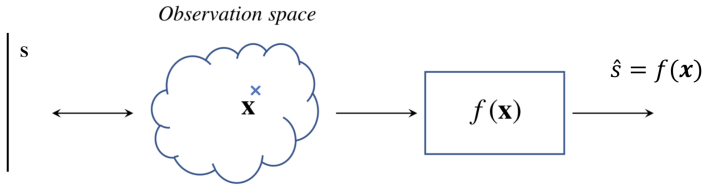


Fig. 1.1 Diagram block of estimation problems.

The estimator's performance is evaluated by how well $\hat{S}$ approximates S, typically using a predefined cost function (or, alternatively, a reward function) that quantifies the estimation error. A well-designed estimator seeks to minimize the expected cost (or to maximize the expected reward). There are two primary tasks in estimation problems:

- Analysis: given an estimator, assess its performance based on statistical criteria.
- Design: find an estimator $f()$ that optimizes a predefined performance goal.

[1] Note that we use capital letters to model the random variables, and we will use lowercase letters to denote an arbitrary realization of them.

1.1.2 Probability model

The statistical relation between the observations and the target variable is described by the **joint** probability density function (pdf), $p_{\mathbf{X},S}(\mathbf{x},s)$, or some distribution related to it. The joint pdf can be factorized as a product of conditional and marginal pdfs:

$$p_{\mathbf{X},S}(\mathbf{x},s) = p_{\mathbf{X}|S}(\mathbf{x}|s) \cdot p_S(s) = p_{S|\mathbf{X}}(s|\mathbf{x}) \cdot p_{\mathbf{X}}(\mathbf{x}) \tag{1.1}$$

In the context of estimation theory, these factors receive specific names:

- The **likelihood** of $S = s$ for observation $\mathbf{x}$, $p_{\mathbf{X}|S}(\mathbf{x}|s)$: it characterizes the generation of observations for each value of the target variable.
- The **prior (or *a priori*) distribution** of S, $p_S(s)$: it describes how much is known (or unknown) about the target variable before observing $\mathbf{X}$.
- The **posterior (or *a posteriori*) distribution** of S given $\mathbf{X} = \mathbf{x}$, $p_{S|\mathbf{X}}(s|\mathbf{x})$: it describes the knowledge (or the uncertainty) about S after observing $\mathbf{X}$.
- The **evidence** or marginal distribution of $\mathbf{X}$, $p_{\mathbf{X}}(\mathbf{x})$.

The information available to design the estimator may depend on the application. In some cases, the likelihood function is known, because it can be related to the physical generative process of the observations. If additionally, a prior distribution is available, the design can be grounded on the posterior distribution $p_{S|\mathbf{X}}(s|\mathbf{x})$, which can be calculated by means of Bayes' Theorem,

$$p_{S|\mathbf{X}}(s|\mathbf{x}) = \frac{p_{\mathbf{X},S}(\mathbf{x},s)}{p_{\mathbf{X}}(\mathbf{x})} = \frac{p_{\mathbf{X}|S}(\mathbf{x}|s)p_S(s)}{\int p_{\mathbf{X}|S}(\mathbf{x}|s')p_S(s')ds'} \tag{1.2}$$

1.1.3 Cost functions for estimation problems

The evaluation and design of an estimator require some objective criteria. In some cases, we will consider that this criterion materializes in the form of a **cost function** whose value we seek to minimize.

A cost function $c(s,\hat{s})$ is any measure of the discrepancy between the target variable and the estimation. It is generally non-negative, $c(s,\hat{s}) \geq 0$, with equality for $s = \hat{s}$. Some cost functions can be expressed as a function of the estimation error $e = s - \hat{s}$ and we will write[2] $c(s,\hat{s}) = c(s-\hat{s}) = c(e)$. Some examples are:

- Square error: $c(e) = e^2$.
- Absolute error: $c(e) = |e|$.
- Relative square error: $c(s,\hat{s}) = \frac{(s-\hat{s})^2}{s^2}$
- Cross Entropy: $c(s,\hat{s}) = -s\ln\hat{s} - (1-s)\ln(1-\hat{s})$, for $s,\hat{s} \in [0,1]$

[2] Note that the cost function is denoted with a lowercase letter, c, because it is a deterministic function, i.e., for fixed values of s and $\hat{s}$ the cost always takes the same value. However, as with the estimation function, the application of that function to random variables will result in another random variable, i.e., $C = c(S,\hat{S})$.

Since the target variable is unknown, the prediction cannot be computed by directly minimizing the cost $c(s,\hat{s})$, and we have to work with expectations. The expected value of the cost is usually referred as the **risk** of an estimator $\hat{s} = f(\mathbf{x})$:

$$R_f = \mathbb{E}\{c(S,\hat{S})\} = \int_{\mathbf{x}}\int_{s} c(s,f(\mathbf{x}))p_{S,\mathbf{X}}(s,\mathbf{x})dsd\mathbf{x} \tag{1.3}$$

By the *Law of Large Numbers*, this is the average cost we can expect from a given estimator, after a large number of predictions.

The **conditional risk** is the conditional mean for a given observation

$$R(\hat{s},\mathbf{x}) = \mathbb{E}\{c(S,\hat{s})|\mathbf{x}\} = \int_{s} c(s,\hat{s})p_{S|\mathbf{X}}(s|\mathbf{x})ds \tag{1.4}$$

Example 1.1 (Evaluation of estimators 1) Given the joint distribution

$$p_{S,X}(s,x) = \begin{cases} \frac{1}{x}, & 0 \le s \le x \text{ and } 0 < x \le 1 \\ 0, & \text{otherwise} \end{cases}, \tag{1.5}$$

consider the estimators $\hat{S}_1 = \frac{1}{2}X$ and $\hat{S}_2 = X$. Which is the best estimator from the point of view of the square error? To find out, we'll calculate the risk (i.e. the mean square error) for both estimators. Knowing that, for any w,

$$\begin{aligned} R &= \mathbb{E}\{(S-wX)^2\} = \int_0^1\int_0^x (s-wx)^2 p_{S,X}(s,x)dsdx = \int_0^1\int_0^x (s-wx)^2\frac{1}{x}dsdx \\ &= \int_0^1 \left(\frac{1}{3}-w+w^2\right)x^2dx = \frac{1}{3}\left(\frac{1}{3}-w+w^2\right) \end{aligned} \tag{1.6}$$

Taking $w = 1/2$ and $w = 1$ we get, respectively, the risks

$$R_1 = \mathbb{E}\left\{(S-\hat{S}_1)^2\right\} = \mathbb{E}\left\{\left(S-\frac{1}{2}X\right)^2\right\} = \frac{1}{3}\left(\frac{1}{3}-\frac{1}{2}+\frac{1}{4}\right) = \frac{1}{36} \tag{1.7}$$

$$R_2 = \mathbb{E}\{(S-\hat{S}_2)^2\} = \mathbb{E}\{(S-X)^2\} = \frac{1}{3}\left(\frac{1}{3}-1+1\right) = \frac{1}{9} \tag{1.8}$$

Therefore, from the point of view of the square error, $\hat{S}_1$ is a better estimator than $\hat{S}_2$.

Example 1.2 (Evaluation of estimators 2) Assume that S is a random variable of mean 0 and variance 1, and X is a noisy observation of S,

$$X = S+T \tag{1.9}$$

where T is a Gaussian random variable, independent of S, with mean 0 variance v. We will compute the risk for the estimator $\hat{S} = X$ and different cost functions. For the square error:

$$\mathbb{E}\{(S-\hat{S})^2\} = \mathbb{E}\{(S-X)^2\} = \mathbb{E}\{T^2\} = v \tag{1.10}$$

For the absolute error

$$\mathbb{E}\{|S-\hat{S}|\} = \mathbb{E}\{|T|\} = \int_{-\infty}^{\infty} |t| \frac{1}{\sqrt{2\pi v}} \exp\left(-\frac{t^2}{2v}\right) dt$$
$$= 2\int_{0}^{\infty} t \frac{1}{\sqrt{2\pi v}} \exp\left(-\frac{t^2}{2v}\right) dt = \sqrt{\frac{2v}{\pi}} \tag{1.11}$$

1.1.4 Bias and Variance

The performance of an estimator $\hat{S} = f(\mathbf{X})$ is often evaluated using two key quantities: *bias* and *variance*. These can be defined in both *conditional* and *unconditional* forms, depending on whether they are measured for a specific target variable or not.

Definition 1.1 (Conditional Bias) The *conditional bias* of an estimator $\hat{S}$ given that $S = s$ is defined as

$$\text{bias}\{\hat{S}|S=s\} = \mathbb{E}\{\hat{S}|S=s\} - s. \tag{1.12}$$

The conditional bias quantifies the expected deviation of the estimate from the true value of the target variable, s.

Theorem 1.1 (Bias-Variance decomposition) *For any estimator, $\hat{S}$ of random variable S, the conditional mean square error can be decomposed as*

$$\mathbb{E}\{(\hat{S}-S)^2|S=s\} = \text{bias}^2\{\hat{S}|S=s\} + \text{var}\{\hat{S}|S=s\} \tag{1.13}$$

Proof Since, for any random variable, V, $\mathbb{E}\{V^2\} = \mathbb{E}\{V\}^2 + \text{var}\{V\}$, taking $V = S - \hat{S}$, and conditioning over $S = s$, we get

$$\mathbb{E}\{(\hat{S}-S)^2|S=s\} = \mathbb{E}\{\hat{S}-S|S=s\}^2 + \text{var}\{\hat{S}-S|S=s\}$$
$$= \text{bias}^2\{\hat{S}|S=s\} + \text{var}\{\hat{S}|S=s\} \tag{1.14}$$

The bias-variance decomposition highlights a major tradeoff in estimation:

- Low-bias estimators tend to have high variance, leading to unstable predictions.
- Low-variance estimators may introduce **bias** to improve stability.

y Note that, in the above expressions, the bias, variance and mean square error are conditioned over a specific value of the target variable. We can extend these definitions to measures defined from unconditional means:

Definition 1.2 (Unconditional Bias) The *unconditional bias* of an estimator $\hat{S}$ of a random variable S is

$$\text{bias}\{\hat{S}\} = \mathbb{E}\{\hat{S}\} - \mathbb{E}\{S\}. \tag{1.15}$$

A similar unconditional bias-variance decomposition can obtained, that is given by

$$\begin{aligned}\mathbb{E}\{(\hat{S}-S)^2\} &= \mathbb{E}\{\hat{S}-S\}^2 + \mathrm{var}\{\hat{S}-S|S=s\} \\ &= \mathrm{bias}^2\{\hat{S}\} + \mathrm{var}\{\hat{S}-S\} \end{aligned} \tag{1.16}$$

If $\mathrm{bias}\{\hat{S}|S=s\}=0$ for all s, the estimator is said to be *conditionally unbiased*. If $\mathrm{bias}\{\hat{S}\}=0$, it is *unbiased in the Bayesian sense*, meaning that its expected value matches the prior mean of S.

1.2 Design of estimators

1.2.1 Maximum Likelihood (ML) estimation

The maximum likelihood estimator (ML) uses the likelihood as the reward function to be maximized:

$$\hat{s}_{\mathrm{ML}} = \underset{s}{\mathrm{argmax}}\, p_{\mathbf{X}|S}(\mathbf{x}|s) = \underset{s}{\mathrm{argmax}} \ln(p_{\mathbf{X}|S}(\mathbf{x}|s)) \tag{1.17}$$

The ML estimator selects the value of the parameter s that maximizes the likelihood of observing $\mathbf{x}$ when $S=s$. Loosely speaking, observing $\mathbf{x}$ when $S=\hat{s}_{\mathrm{ML}}$ is less unexpected than if S takes any other value. Note that $p_{\mathbf{X}|S}(\mathbf{x}|s)$, which is a density function over random variable $\mathbf{X}$, is not maximized with respect to $\mathbf{x}$, but s.

Example 1.3 (ML Estimation)

We want to compute the ML estimate of a random variable S given observation X. The likelihood function is given by

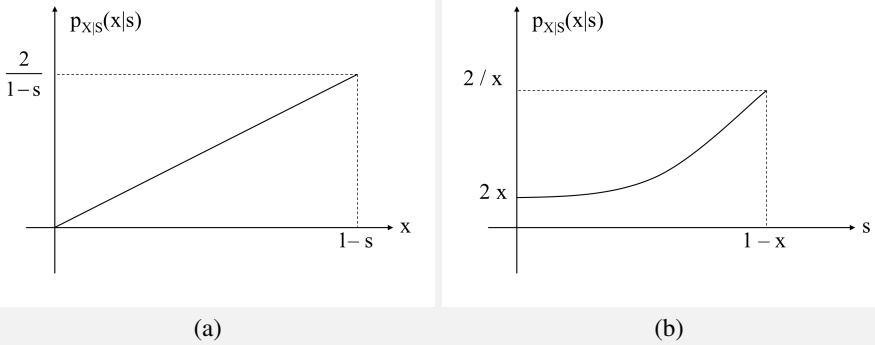
$$p_{X|S}(x|s) = \frac{2x}{(1-s)^2}, \qquad 0 \le x \le 1-s, \quad 0 \le s \le 1 \tag{1.18}$$

and it is represented in Figure 1.2(a) as a function of x, for a specific value of s. Note that the likelihood function is a conditional density function of the random variable X. However, since we aim to estimate S, it may be more useful to represent the likelihood as a function of s, for a given observation x, as shown in Fig.1.2(b),[a]. This representation suggests that the ML estimator is

$$\hat{s}_{\mathrm{ML}} = 1-x$$

which can be corroborated by inspecting (1.18), which shows that the likelihood increases with s. If we apply the estimation function to random variable X instead of a specific value, we have

$$\hat{S}_{\mathrm{ML}} = 1-X$$



(a) (b)

Fig. 1.2 Representation of the likelihood distribution of the example 1.3 as a function of x and s.

[a] Note that the integral with respect to s of $p_{X|S}(x|s)$ will not generally be equal to one, since the likelihood function is not a probability density function of the target variable S.

Note that the second equality in (1.17) states that the maximization of the likelihood is equivalent to the maximization of its logarithm (the **log-likelihood** function). Since the logarithm function is strictly increasing, $p_{\mathbf{X}|S}(\mathbf{x}|s_1) > p_{\mathbf{X}|S}(\mathbf{x}|s_2)$ implies $\ln(p_{\mathbf{X}|S}(\mathbf{x}|s_1)) > \ln(p_{\mathbf{X}|S}(\mathbf{x}|s_2)))$ and, thus, the logarithm does not alter the outcome of the maximization. The logarithm is useful to simplify the optimization when the likelihood is a product of several factors or an exponential function, as it will transform products into sums and an exponential into the exponents.

Note that the maximum likelihood does not need any probability model about the target variable, S, which is treated as a deterministic parameter.

1.2.2 Maximum a posteriori (MAP) estimation

We define the maximum a posteriori (MAP) estimator as the mode of the posterior distribution, that is

$$\hat{s}_{\text{MAP}} = \underset{s}{\text{argmax}}\, p_{S|\mathbf{X}}(s|\mathbf{x}) = \underset{s}{\text{argmax}} \ln(p_{S|\mathbf{X}}(s|\mathbf{x})) \tag{1.19}$$

Using the definition of conditional pdf and Bayes' rule, it is easy to see that the MAP estimator can be altenatively computed as

$$\hat{s}_{\text{MAP}} = \underset{s}{\text{argmax}}\, p_{S,\mathbf{X}}(s,\mathbf{x}) \tag{1.20}$$

$$= \underset{s}{\text{argmax}}\{p_{\mathbf{X}|S}(\mathbf{x}|s)p_S(s)\} \tag{1.21}$$

Eq. (1.21) demonstrates that the MAP estimator seeks to maximize the likelihood function, modulated by the prior distribution. This integration of the prior distribution allows the MAP estimator to incorporate existing knowledge or assumptions about the target S before observing the data, $\mathbf{x}$. When the prior distribution is uniform across the domain of S, the distinction between the MAP and ML estimators vanishes, as the MAP estimator effectively reduces to the ML estimator.

However, when the prior distribution is not uniform, the MAP estimator is biased towards values of the target variable with higher prior probabilities. This shift illustrates the MAP estimator's sensitivity to prior knowledge.

Beyond their mathematical formulations, the MAP and ML estimators embody fundamentally different inference philosophies. The ML estimator optimizes the likelihood function, which models the probability of the observed data under various values of the target, treating s as a fixed but unknown parameter. This approach aligns with the **frequentist paradigm**, which interprets probability as the long-run frequency of events and does not incorporate prior information about S.

Conversely, the MAP estimator embraces a **Bayesian framework**, treating the target as a random variable. This perspective allows the incorporation of prior knowledge or beliefs about S through the prior distribution, $p_S(s)$, and the posterior distribution, $p_{S|\mathbf{X}}(s|\mathbf{x})$, updates this knowledge based on new evidence from the data. The Bayesian approach, therefore, provides a probabilistic framework for updating beliefs about uncertain parameters in light of new data.

Example 1.4 (MAP estimation) Given the posterior density

$$p_{S|X}(s|x) = \frac{1}{x^2} s \exp\left(-\frac{s}{x}\right), \qquad x \geq 0, \quad s \geq 0 \tag{1.22}$$

the MAP estimator can be computed by maximizing

$$\ln(p_{S|X}(s|x)) = -2\ln(x) + \ln(s) - \frac{s}{x}, \qquad x \geq 0, \quad s \geq 0, \tag{1.23}$$

Since $\ln(p_{S|X}(s|x))$ tends to $-\infty$ around $s = 0$ and $s = \infty$, its maximum must be at some intermediate point with zero derivative:

$$\frac{\partial}{\partial s} \ln p_{S|X}(s|x)\Big|_{s=\hat{s}_{\text{MAP}}} = \frac{1}{\hat{s}_{\text{MAP}}} - \frac{1}{x} = 0, \qquad x \geq 0, \quad s \geq 0. \tag{1.24}$$

Therefore,

$$\hat{s}_{\text{MAP}} = x \tag{1.25}$$

1.2.3 Minimum risk estimators

On major drawback of the MAP estimator is that a high posterior density at a given point does not necessarily mean that the estimated value is likely to be close to the true parameter, as the example in Fig. 1.3 shows, where the MAP estimate lies in a region with low probability. This motivates the application of other Bayesian estimators that take into account the cost of large deviations from the true parameter.

The Bayesian estimator for a cost function $c(S,\hat{S})$ is the one minimizing the risk (1.3). Applying the total expectation theorem, the risk of estimator $\hat{s} = f(\mathbf{x})$ can be expressed as

$$R_f = \mathbb{E}\{c(S,\hat{S})\} = \int_{\mathbf{x}} \mathbb{E}\{c(S,\hat{s})|\mathbf{X} = \mathbf{x}\} p_{\mathbf{X}}(\mathbf{x}) \mathbf{dx} \tag{1.26}$$

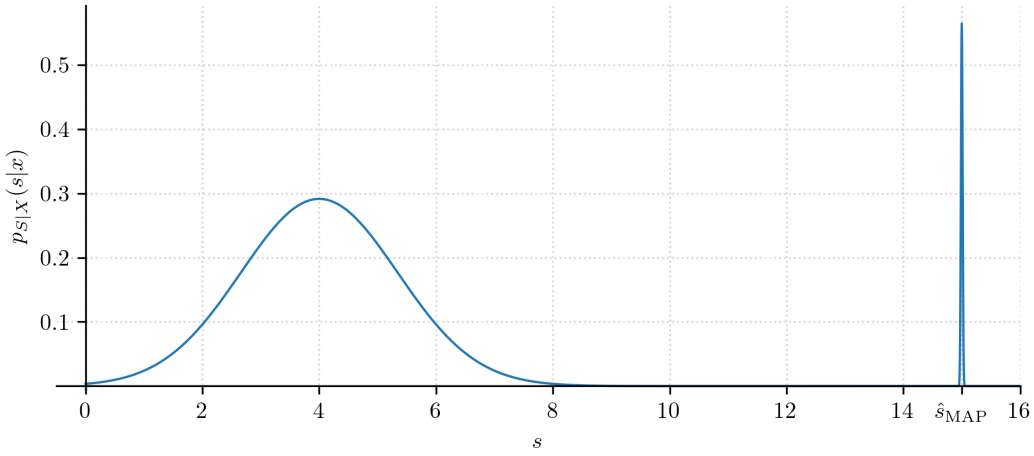


Fig. 1.3 A visual example of a posterior distribution, where the MAP estimate can be far from the true value of the target parameter with high probability.

where $\hat{s} = f(\mathbf{x})$. That is, the risk is the integral of the conditional risk, and, thus, the estimator minimizing R_f should minimize the conditional risk for each observation $\mathbf{x}$,

$$\hat{s}^* = \underset{\hat{s}}{\operatorname{argmin}}\ \mathbb{E}\{c(S,\hat{s})|\mathbf{X}=\mathbf{x}\} \tag{1.27}$$

Example 1.5 (Calculation of a minimum mean square error estimator) Following the example 1.1, we can calculate the posterior distribution of S through

$$p_{S|X}(s|x) = \frac{p_{S,X}(s,x)}{p_X(x)}. \tag{1.28}$$

Knowing that

$$p_X(x) = \int_0^1 p_{S,X}(s,x)ds = \int_0^x \frac{1}{x}ds = 1, \qquad 0 \le x \le 1 \tag{1.29}$$

we obtain

$$p_{S|X}(s|x) = \begin{cases} \frac{1}{x}, & 0 < s < x < 1 \\ 0, & \text{otherwise} \end{cases} \tag{1.30}$$

The conditional risk will be given by

$$\begin{aligned} \mathbb{E}\{c(S,\hat{s})|X=x\} &= \mathbb{E}\{(S-\hat{s})^2|X=x\} \\ &= \int_0^1 (s-\hat{s})^2 p_{S|X}(s|x)ds \\ &= \frac{1}{x}\int_0^x (s-\hat{s})^2 ds = \frac{1}{x}\left(\frac{(x-\hat{s})^3}{3} + \frac{\hat{s}^3}{3}\right) \\ &= \frac{1}{3}x^2 - \hat{s}x + \hat{s}^2. \end{aligned} \tag{1.31}$$

As a function of $\hat{s}$, conditional risk is a second-degree polynomial, whose minimum can be calculated through differentiation. Since

$$\frac{d}{d\hat{s}}\mathbb{E}\{c(S,\hat{s})|X=x\} = -x+2\hat{s}, \tag{1.32}$$

the Bayesian estimator associated with the square error is

$$\hat{s}^* = \frac{1}{2}x, \tag{1.33}$$

which matches the estimator $\hat{S}_1$ from the example 1.1. Therefore, $\hat{S}_1$ is the best possible estimator from the point of view of the mean square error.

1.3 Common Bayesian estimators

In this section we derive some of the most widely used Bayesian estimators, starting from different choices of the loss function.

1.3.1 Minimum Mean Squared Error estimator (MSE)

The minimum mean squared error (MSE) estimator is the Bayesian estimator associated with the cost function $c(e) = e^2 = (s-\hat{s})^2$, and therefore is given by

$$\begin{aligned}\hat{s}_{\text{MSE}} &= \underset{\hat{s}}{\operatorname{argmin}}\ \mathbb{E}\{c(S,\hat{s})|\mathbf{X}=\mathbf{x}\} \\ &= \underset{\hat{s}}{\operatorname{argmin}}\ \mathbb{E}\{(S-\hat{s})^2|\mathbf{X}=\mathbf{x}\}\end{aligned} \tag{1.34}$$

Figure 1.4 illustrates the minimum MSE estimation problem. The risk can be obtained by integrating (with respect to s) the product of the square error and the posterior pdf of S. The argument for minimization is $\hat{s}$, which allows shifting the graph corresponding to the cost function (represented with discontinuous stroke) so that the result of that integral is minimal.

For the square error, the conditional risk in (1.34) becomes

$$\mathbb{E}\{(S-\hat{s})^2|\mathbf{X}=\mathbf{x}\} = \mathbb{E}\{S^2|\mathbf{X}=\mathbf{x}\} - 2\mathbb{E}\{S|\mathbf{X}=\mathbf{x}\}\hat{s}+\hat{s}^2 \tag{1.35}$$

This is a second-degree polynomial that can be minimized by differentiation to get

$$\hat{s}_{\text{MSE}} = \mathbb{E}\{S|\mathbf{X}=\mathbf{x}\} = \int s\cdot p_{S|\mathbf{X}}(s|x)ds \tag{1.36}$$

In other words, the minimum MSE estimator of S is the posterior mean of S given $\mathbf{X}$.

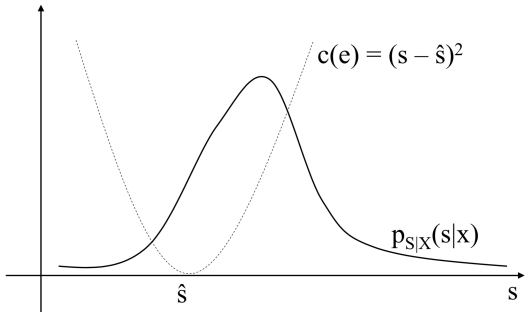


Fig. 1.4 Graphical representation of the process of calculating the posterior mean for a generic value $\hat{s}$.

Example 1.6 (MSE estimator) According to (1.36), minimum mean squared error estimator obtained in 1.1 can alternatively be derived as follows

$$\hat{s}_{\text{MSE}} = \int_0^1 s p_{S|X}(s|x) ds = \int_0^x \frac{s}{x} ds = \frac{1}{2} x \tag{1.37}$$

which is consistent with (1.33).

1.3.2 Minimum Mean Absolute Deviation Estimator (MAD)

When the cost is the absolute deviation, $c(e) = |e| = |s - \hat{s}|$, the minimizer of the conditional risk (which we will refer to as the Mean Absolute Deviation, or MAD, estimator), is given by

$$\hat{s}_{\text{MAD}} = \underset{\hat{s}}{\text{argmin}}\ \mathbb{E}\{|S - \hat{s}|\ |\mathbf{X} = \mathbf{x}\} = \tag{1.38}$$

$$= \underset{\hat{s}}{\text{argmin}} \int_s |s - \hat{s}|\ p_{S|\mathbf{X}}(s|\mathbf{x}) ds \tag{1.39}$$

The computation of the conditional risk is illustrated in Fig. 1.5. This representation also suggests the convenience of splitting the integral into two parts corresponding to the two slopes of the cost function:

$$\mathbb{E}\{|S - \hat{s}|\ |\mathbf{X} = \mathbf{x}\} = \int_{-\infty}^{\hat{s}} (\hat{s} - s)\ p_{S|\mathbf{X}}(s|\mathbf{x}) ds + \int_{\hat{s}}^{\infty} (s - \hat{s})\ p_{S|\mathbf{X}}(s|\mathbf{x}) ds \tag{1.40}$$

$$= \hat{s} \left[\int_{-\infty}^{\hat{s}} p_{S|\mathbf{X}}(s|\mathbf{x}) ds - \int_{\hat{s}}^{\infty} p_{S|\mathbf{X}}(s|\mathbf{x}) ds \right] + \int_{\hat{s}}^{\infty} s\ p_{S|\mathbf{X}}(s|\mathbf{x}) ds - \int_{-\infty}^{\hat{s}} s\ p_{S|\mathbf{X}}(s|\mathbf{x}) ds \tag{1.41}$$

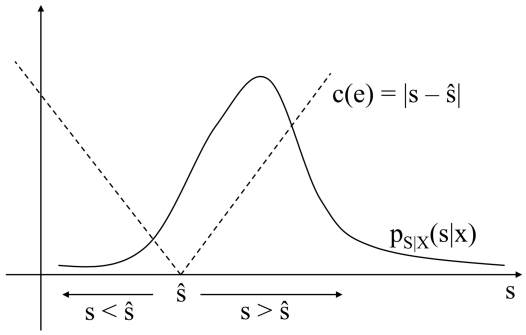


Fig. 1.5 Calculation of the posterior mean absolute error for a generic value $\hat{s}$.

Applying the fundamental theorem of calculus[3] we can write[4]

$$\frac{d\mathbb{E}\{|S-\hat{s}|\ |\mathbf{X}=\mathbf{x}\}}{d\hat{s}} = 2F_{S|\mathbf{X}}(\hat{s}|\mathbf{x}) - 1 \tag{1.42}$$

where $F_{S|\mathbf{X}}(s|\mathbf{x})$ is the posterior cumulative distribution function of S given $\mathbf{X}$. Since this derivative must vanish at the minimum, we get $F_{S|\mathbf{X}}(\hat{s}_{\text{MAD}}|\mathbf{x}) = 1/2$. In other words, the minimum MAD estimator is given by the median of $p_{S|\mathbf{X}}(s|\mathbf{x})$:

$$\hat{s}_{\text{MAD}} = \text{median}\{S|\mathbf{X}=\mathbf{x}\} \tag{1.43}$$

Remember that the median of a distribution is the point that splits its domain into two sets with the same probability, so the minimum MAD estimator satisfies

$$P\{S > \hat{s}_{\text{MAD}}|\mathbf{x}\} = P\{S < \hat{s}_{\text{MAD}}|\mathbf{x}\} \tag{1.44}$$

In practice, this can be computed as the solution of

$$\int_{-\infty}^{\hat{s}_{\text{MAD}}} p_{S|\mathbf{X}}(s|\mathbf{x})ds = \frac{1}{2} \tag{1.45}$$

Example 1.7 (MAD Estimator)
In the scenario of the example 1.1, the posterior distribution of S given X is uniform between 0 and x, the median of which is $x/2$. Thus,

$$\hat{s}_{\text{MAD}} = \frac{1}{2}x \tag{1.46}$$

[3] $\frac{d}{dx}\int_{t_0}^{x} g(t)dt = g(x)$.

[4] Strictly speaking, it is assumed here that the posterior distribution is absolutely continuous, that is, $p_{S|X}(s|x)$ does not contain singularities (i.e. points with nonzero probability mass).

Note that, in this case, the MAD estimator matches the MSE obtained at (1.33). This is a consequence of the symmetry of the a posterior distribution.

1.4 Estimation with constrains

1.4.1 General principles

Occasionally, it might be beneficial to prescribe a specific parametric form for the estimator, denoted as $\hat{S} = f_{\mathbf{w}}(\mathbf{X})$, where $\mathbf{w}$ represents a vector of parameters. For instance, in scenarios involving two observations, $\mathbf{X} = [X_1, X_2]^T$, a design constraint might necessitate limiting the search for an estimator to the family of quadratic estimators, characterized by $\hat{S} = w_0 + w_1 X_1^2 + w_2 X_2^2$. In such situations, the task of designing an estimator consists of identifying the optimal parameter vector $\mathbf{w}^*$ that minimizes the risk, while adhering to the specified constraints on the estimator structure:

$$\begin{aligned}\mathbf{w}^* &= \underset{\mathbf{w}}{\operatorname{argmin}}\, \mathbb{E}\{c(S,\hat{S})\} = \underset{\mathbf{w}}{\operatorname{argmin}}\, \mathbb{E}\{c(S, f_{\mathbf{w}}(\mathbf{X}))\} \\ &= \underset{\mathbf{w}}{\operatorname{argmin}} \int_{\mathbf{x}}\int_{s} c(s, f_{\mathbf{w}}(\mathbf{x})) p_{S,\mathbf{X}}(s,\mathbf{x}) ds d\mathbf{x}. \end{aligned} \tag{1.47}$$

Restricting the estimator to a specific analytic form typically results in a higher risk than what could be achieved with a Bayesian estimator tailored to the same cost function. The exception to this rule occurs when the imposed constraints align with the optimal estimator form, essentially when the Bayesian estimator inherently fits within the specified constraints. Despite potentially incurring higher costs, practical considerations may justify opting for such constrained estimators, such as simplification in design or implementation. An exploration of this concept is presented in Section 1.4.2, focusing on linear estimators that achieve minimum MSE.

Example 1.8 (Estimator with Constraints)
Continuing the example 1.5, we want to calculate the minimum MSE estimator with the form $\hat{s} = wx^2$. Since the conditional risk is given by (1.31), the risk can be computed as

$$\begin{aligned}\mathbb{E}\{c(S,\hat{S})\} &= \int_x \mathbb{E}\{c(S,\hat{s})|X=x\}\; p_X(x)dx \\ &= \int_x \left(\frac{1}{3}x^2 - \hat{s}x + \hat{s}^2\right) p_X(x)dx \end{aligned} \tag{1.48}$$

Forcing $\hat{s} = wx^2$ and taking into account that $p_X(x) = 1$ for $0 < x < 1$, we get the MSE as a function of w.

$$\mathbb{E}\{c(S,wX^2)\} = \int_0^1 \left(\frac{1}{3}x^2 - wx^3 + w^2x^4\right) dx \tag{1.49}$$

$$= \frac{1}{9} - \frac{1}{4}w + \frac{1}{5}w^2 \tag{1.50}$$

Since the risk is a quadratic funtion of w, it can be optimized by differentiation, to get

$$w^* = \frac{5}{8}, \tag{1.51}$$

and therefore the estimator is $\hat{s} = \frac{5}{8}x^2$.

1.4.2 Linear (in the parameters) estimation of minimum MSE

In this section we will focus on the study of estimators that are a linear combination of variables related to the observations, using the minimization of the MSE as a design criterion. More specifically, we will consider estimators given by the general expression

$$\hat{S} = \mathbf{w}^\intercal \mathbf{Z} \tag{1.52}$$

where

$$\mathbf{Z} = \phi(\mathbf{X}) \tag{1.53}$$

is some known transformation of the observations. The nature of this transformation may depend on the application, but there are some cases of particular interest:

- **Linear estimation**: in this case, ϕ is the identity function, so that $\mathbf{Z} = \mathbf{X}$, and the estimator a linear function of the observations

$$\hat{S} = w_0 X_0 + w_1 X_1 + \cdots + w_{N-1} X_{N-1} \tag{1.54}$$

- **Linear estimation with independent term**: in this case, $\mathbf{Z} = (1, \mathbf{X}^\intercal)^\intercal$ and

$$\hat{S} = w_0 + w_1 X_0 + \cdots + w_N X_{N-1} \tag{1.55}$$

- **Polynomial estimation**: the components of $\mathbf{Z}$ are monomials of the observational variables. For instance, for a scalar observation $X \in \mathbb{R}$, we can take $\mathbf{Z} = (1, X, X^2, \ldots X^M)$, for some $M > 1$, and the estimation becomes an M-degree polynomial of the observation.

$$\hat{S} = w_0 + w_1 X + w_2 X^2 \cdots + w_M X^M \tag{1.56}$$

Note that in general, $\mathbf{X}$ and $\mathbf{Z}$ may have different dimensions. Note, also, that, despite the estimator may be a non-linear function of the observation, all estimators given by (1.52) are linear functions of the parameters. For this reason, we will refer to them as estimators that are *linear in the parameters*.

By imposing a restriction on the analytic form of the estimator, linear-in-the-parameter estimators will generally obtain lower performance than the optimal Bayesian estimator. However, the interest in linear estimators is justified by their simplicity and ease of design. As we shall see, the linear estimator of minimum MSE depends exclusively on first and second-order statistical moments (means and covariances) associated with the target variable and the transformed observation, $\mathbf{Z}$.

1.4.2.1 Minimization of the mean squared error.

We will consider as design criteria the squared error, $c(e) = (s - \hat{s})^2$, so the optimal weight vector will be the one that minimizes the MSE risk:

$$\mathbf{w}^* = \underset{\mathbf{w}}{\operatorname{argmin}}\ \mathbb{E}\{(S - \hat{S})^2\} = \underset{\mathbf{w}}{\operatorname{argmin}}\ \mathbb{E}\{(S - \mathbf{w}^\intercal \mathbf{Z})^2\} \tag{1.57}$$

and we will refer to the estimator associated with the weight vector as $\hat{S}_{\text{LMSE}}$:

$$\hat{S}_{\text{LMSE}} = \mathbf{w}^{*T}\mathbf{Z}$$

The MSE can be expanded as

$$\begin{aligned} MSE &= \mathbb{E}\{(S - \hat{S})^2\} = \mathbb{E}\{(S - \mathbf{w}^\intercal \mathbf{Z})^2\} \\ &= \mathbb{E}\{S^2\} - 2\mathbb{E}\{\mathbf{w}^\intercal \mathbf{Z} S\} + \mathbb{E}\{(\mathbf{w}^\intercal \mathbf{Z})^2\} \\ &= \mathbb{E}\{S^2\} - 2\mathbb{E}\{S\mathbf{Z}\}^\intercal \mathbf{w} + \mathbf{w}^\intercal \mathbb{E}\{\mathbf{Z}\mathbf{Z}^\intercal\}\mathbf{w} \\ &= \mathbb{E}\{S^2\} - 2\mathbf{r}_{S\mathbf{Z}}^\intercal \mathbf{w} + \mathbf{w}^\intercal \mathbf{R}_{\mathbf{Z}}\mathbf{w} \end{aligned} \tag{1.58}$$

where

- $\mathbf{r}_{S\mathbf{Z}} = \mathbb{E}\{S\mathbf{Z}\}$ is the **cross-correlation vector** between S and $\mathbf{Z}$
- $\mathbf{R}_{\mathbf{Z}} = \mathbb{E}\{\mathbf{Z}\mathbf{Z}^\intercal\}$ is the **autocorrelation matrix** of $\mathbf{Z}$.

Thus, the MSE is a second-degree polynomial in $\mathbf{w}$. This is illustrated in Figure 1.6, which depicts the error surface in a case with two observations. Being the function to minimize quadratic in weights (minimization argument), the error surface will take the form of a N dimensional paraboloid.

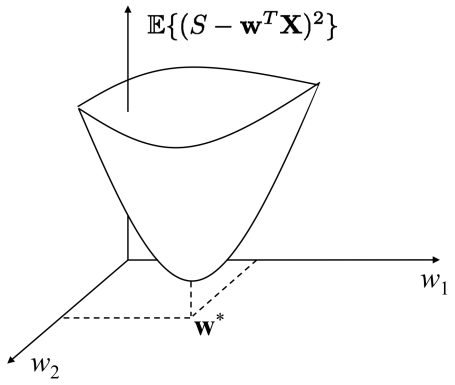


Fig. 1.6 Surface of the MSE for the linear estimator with $\mathbf{Z} = \mathbf{X}$, as a function of the parameters.

Since the MSE is non-negative, it is guaranteed to be a convex function of $\mathbf{w}$ and, thus, its minimum must be located at a point with zero gradient[5] (with respect to $\mathbf{w}$):

$$\nabla_{\mathbf{w}} MSE|_{\mathbf{w}=\mathbf{w}^*} = -2\mathbf{r}_{S\mathbf{Z}} + 2\mathbf{R}_{\mathbf{Z}}\mathbf{w}^* = \mathbf{0} \tag{1.59}$$

therefore, the optimal weight vector is any solution of

$$\mathbf{R}_{\mathbf{Z}}\mathbf{w}^* = \mathbf{r}_{S\mathbf{Z}} \tag{1.60}$$

If the autocorrelation matrix is invertible, we get

$$\boxed{\mathbf{w}^* = \mathbf{R}_{\mathbf{Z}}^{-1}\mathbf{r}_{S\mathbf{Z}}} \tag{1.61}$$

1.4.2.2 Alternative expression for the linear case

We can obtain an alternative expression for the optimal weights in the case of the linear estimator with the independent term in (1.55), that can be written as

$$\hat{S} = w_0 + \mathbf{w}_{1:}^{\intercal}\mathbf{X} \tag{1.62}$$

where $\mathbf{w}_{1:}$ is the weight vector after removing the first component, w_0. To do so, we can re-express (1.60) as the equivalent pair of equations

$$\mathbb{E}\{w_0^* + \mathbf{w}_{1:}^{*\intercal}\mathbf{X}\} = \mathbb{E}\{S\} \tag{1.63}$$

$$\mathbb{E}\{\mathbf{X}\}w_0^* + \mathbb{E}\{\mathbf{X}\mathbf{X}^{\intercal}\}\mathbf{w}_{1:}^* = \mathbb{E}\{S\mathbf{X}\} \tag{1.64}$$

that is

$$w_0^* = m_S - \mathbf{w}_{1:}^{*\intercal}\mathbf{m}_{\mathbf{x}} \tag{1.65}$$

$$\mathbb{E}\{\mathbf{X}\mathbf{X}^{\intercal}\}\mathbf{w}_{1:}^* = \mathbb{E}\{S\mathbf{X}\} - w_0^*\mathbf{m}_{\mathbf{X}} \tag{1.66}$$

where $m_S = \mathbb{E}\{S\}$, $m_{\mathbf{X}} = \mathbb{E}\{\mathbf{X}\}$. Now using the expressions that relate the correlation and covariance of two variables:

$$\mathbb{E}\{S\mathbf{X}\} = \mathbf{v}_{S\mathbf{X}} + m_S\mathbf{m}_{\mathbf{X}} \tag{1.67}$$

$$\mathbb{E}\{\mathbf{X}\mathbf{X}^{\intercal}\} = \mathbf{V}_{\mathbf{X}} + \mathbf{m}_{\mathbf{X}}\mathbf{m}_{\mathbf{X}}^{\intercal} \tag{1.68}$$

we get

$$w_0^* = m_S - \mathbf{w}_{1:}^{*\intercal}\mathbf{m}_{\mathbf{x}} \tag{1.69}$$

$$\mathbf{v}_{S\mathbf{X}} + m_S\mathbf{m}_{\mathbf{X}} = (\mathbf{V}_{\mathbf{X}} + \mathbf{m}_{\mathbf{X}}\mathbf{m}_{\mathbf{X}}^{T})\mathbf{w}_{1:}^* + w_0^*\mathbf{m}_{\mathbf{X}} \tag{1.70}$$

Solving these equations for w_0^* and $w_{1:}^*$ we get

[5] The gradient of a function scale $f(\mathbf{w})$ with respect to the vector $\mathbf{w}$ is defined as a vector formed by the derivatives of the function with respect to each one of the components of $\mathbf{w}$: $\nabla_{\mathbf{w}} f(\mathbf{w}) = \left[\frac{\partial f}{\partial w_1}, \ldots \frac{\partial f}{\partial w_N}\right]^T$.

$$w_0^* = m_S - \mathbf{w}_{1:}^{*\,T}\mathbf{m}_{\mathbf{x}} \tag{1.71}$$

$$\mathbf{w}_{1:}^* = \mathbf{V}_{\mathbf{X}}^{-1}\,\mathbf{v}_{S,\mathbf{X}} \tag{1.72}$$

The optimal estimator is, thus,

$$\hat{S}_{\text{LMSE}} = m_S + \mathbf{v}_{S\mathbf{X}}^{\intercal}\mathbf{V}_{\mathbf{X}}(\mathbf{X} - \mathbf{m}_{\mathbf{X}}) \tag{1.73}$$

The bias term w_0 compensates for differences between the means of the target variable and the observations. Therefore, when all the variables involved have zero mean, $w_0^* = 0$, and the estimator is purely linear in the observations.

1.4.2.3 Minimum squared mean error

The minimum MSE can be computed by replacing the optimal weights (1.61) into (1.58)

$$\begin{aligned}\text{MSE}^* &= \mathbb{E}\{S^2\} - 2\mathbf{r}_{S\mathbf{Z}}^{\intercal}\mathbf{R}_{\mathbf{Z}}^{-1}\mathbf{r}_{S\mathbf{Z}} + (\mathbf{R}_{\mathbf{Z}}^{-1}\mathbf{r}_{S\mathbf{Z}})^{\intercal}\mathbf{R}_{\mathbf{Z}}\mathbf{R}_{\mathbf{Z}}^{-1}\mathbf{r}_{S\mathbf{Z}} \\ &= \mathbb{E}\{S^2\} - \mathbf{r}_{S\mathbf{Z}}^{\intercal}\mathbf{R}_{\mathbf{Z}}^{-1}\mathbf{r}_{S\mathbf{Z}}\end{aligned} \tag{1.74}$$

For the linear estimator (1.73)

$$\text{MSE}^* = v_S - \mathbf{w}^{*\,T}\mathbf{v}_{S\mathbf{X}} \tag{1.75}$$

1.5 Estimation with Gaussian distributions

In this section, we delve into the estimation of random variables within the context where the combined distribution of all involved variables (the target variable along with the observational variables) is a multidimensional Gaussian. This scenario is particularly significant due to the prevalent occurrence of these distributions in signal processing, communications and various other fields.

When the joint distribution $p_{S,\mathbf{X}}(s,\mathbf{x})$ is Gaussian, all marginal and conditional distributions retain a Gaussian form. In particular, the posterior distribution, $p_{S|\mathbf{X}}(s|\mathbf{x})$ is Gaussian. Since the mean, median, and mode of the Gaussian distribution align, $\hat{s}_{\text{MSE}} = \hat{s}_{\text{MAD}} = \hat{s}_{\text{MAP}}$. Thus, our discussion in this section will primarily concentrate on deriving the estimator that minimizes the MSE.

Besides, we will demonstrate that the minimum MSE estimator and, consequently, the MAP and MAD estimators are linear, which will allow us to use the results shown in the previous section for minimum MSE estimation.

1.5.1 One dimensional variables

Assume that the joint distribution of X and S is Gaussian:

$$p_{S,X}(s,x) \sim G\left(\begin{bmatrix} s \\ x \end{bmatrix}, \begin{bmatrix} v_X & \rho \\ \rho & v_S \end{bmatrix}\right) \tag{1.76}$$

where ρ is the covariance between the two random variables. From the joint distribution, we can obtain any other distribution involving the variables S and X; specifically, the posterior distribution can be obtained as:

$$p_{S|X}(s|x) = \frac{p_{S,X}(s,x)}{p_X(x)} = \frac{\frac{1}{2\pi\sqrt{v_X v_S - \rho^2}} \exp\left[-\frac{1}{2(v_X v_S - \rho^2)} \begin{bmatrix} s \\ x \end{bmatrix}^{\mathsf{T}} \begin{bmatrix} v_X & -\rho \\ -\rho & v_S \end{bmatrix} \begin{bmatrix} s \\ x \end{bmatrix}\right]}{\frac{1}{\sqrt{2\pi v_X}} \exp\left[-\frac{x^2}{2v_X}\right]} \tag{1.77}$$

where it has been necessary to calculate the inverse of the covariance matrix of S and X.

Noting that, as a function of s, $p_{S|X}(s|x)$ differs from $p_{S,X}(s,x)$ in the scale factor $p_X(x)$, which does not depend on s, $p_{S|X}(s|x)$ should be a Gaussian pdf too. Therefore, it must be writable as

$$p_{S|X}(s|x) = \frac{1}{\sqrt{2\pi v_{S|X}}} \exp\left[-\frac{(s - m_{S|X})^2}{2v_{S|X}}\right] \tag{1.78}$$

where $m_{S|X}$ and $v_{S|X}$ are the posterior mean and variance, respectively, to be determined.

Joining (1.77) and (1.78), we can write

$$\frac{2\pi\sqrt{v_X v_S - \rho^2}}{\sqrt{2\pi v_{S|X}}\sqrt{2\pi v_X}} \exp\left[-\frac{(s - m_{S|X})^2}{2v_{S|X}} + \frac{1}{2(v_X v_S - \rho^2)} \begin{bmatrix} s \\ x \end{bmatrix}^{T} \begin{bmatrix} v_X & -\rho \\ -\rho & v_S \end{bmatrix} \begin{bmatrix} s \\ x \end{bmatrix} - \frac{x^2}{2v_X}\right] = 1 \tag{1.79}$$

which can be simplified to

$$\frac{\sqrt{v_X v_S - \rho^2}}{\sqrt{v_{S|X} v_X}} \exp\left[-\frac{s^2 - 2m_{S|X}s + m_{S|X}^2}{2v_{S|X}} + \frac{v_X s^2 - 2\rho x s + v_S x^2}{2(v_X v_S - \rho^2)} - \frac{x^2}{2v_X}\right] = 1 \tag{1.80}$$

Note that the equation above must be satisfied for any $s \in \mathbb{R}$. Since the right-hand side is constant, and the exponent on the left-hand side is a polynomial function of s, the coefficients multiplying s^2 and s must be zero. Thus

$$\frac{m_{S|X}}{v_{S|X}} = \frac{\rho x}{v_X v_S - \rho^2} \tag{1.81}$$

$$\frac{1}{v_{S|X}} = \frac{v_X}{v_X v_S - \rho^2} \tag{1.82}$$

From (1.82) we get the posterior variance

$$v_{S|X} = v_S - \frac{\rho^2}{v_X} \tag{1.83}$$

and, replacing (1.83) into (1.81) we get the posterior mean, which is the minimum MSE estimate

$$\hat{s}_{\text{MSE}} = \hat{s}_{\text{MAD}} = \hat{s}_{\text{MAP}} = m_{S|X} = \frac{\rho}{v_X} x \tag{1.84}$$

Note that the estimator is a linear function of the observation.

Exercise 1.1 Generalize the above result for the case where the variables S and X have (non-zero) means m_S and m_X, respectively. Demonstrate that in such a case, the estimator is

$$\hat{s}_{\text{MSE}} = m_S + \frac{\rho}{v_X}(x - m_X) \tag{1.85}$$

Example 1.9 (Estimation of a Gaussian signal contaminated by Gaussian noise)

In this example, we will consider the case in which the observation is the sum of the target variable and an independent noise component: $X = S + R$. Both the target and the noise are zero-mean Gaussian random variables with variances v_S and v_R, respectively.

Figure (1.7) represents the situation described for a case with $v_S < v_R$.

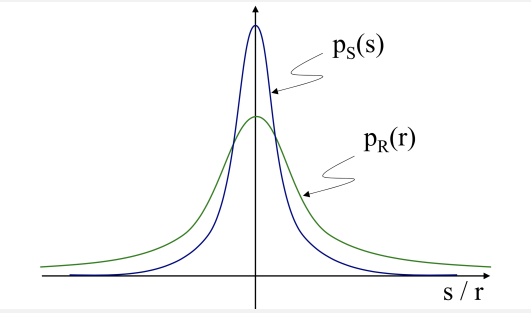


Fig. 1.7 Estimation of Gaussian random variable S contaminated by Gaussian noise R.

According to (1.84), for the resolution of the problem, we must find the variance of X and the covariance between S and X (ρ). The variance v_X is obtained simply as the sum of v_S and v_R because both are independent variables. For the covariance calculation,

$$\rho = \mathbb{E}\{(X - m_X)(S - m_S)\} = \mathbb{E}\{X\,S\} = \mathbb{E}\{(S+R)S\} = \mathbb{E}\{S^2\} + \mathbb{E}\{S\,R\} = v_S \tag{1.86}$$

where independence of S and R has been used, and the fact that all variables (including X) have zero means.

Replacing these results in (1.84) we get

$$\hat{s}_{\text{MSE}} = \frac{v_S}{v_S + v_R} x \tag{1.87}$$

This result can be interpreted quite intuitively: when the variance of the noise is much lower than that of the signal (high Signal to Noise Ratio (SNR), $v_S \gg v_R$) we get $\hat{s}_{\text{MSE}} \to x$, which makes sense since the effect of the noise component, in this case, is not very significant; on the contrary, when the SNR is very small ($v_S \ll v_R$), the observation barely provides information about the S value in each experiment, so the estimator keeps the mean value of the signal component, $\hat{s}_{\text{MSE}} \to 0$.

1.5.2 Multidimensional variables

In a general multidimensional case, $\mathbf{S}$ and $\mathbf{X}$ can be random vectors of dimensions N and M, respectively, with joint Gaussian distribution.

$$p_{\mathbf{S},\mathbf{X}}(\mathbf{s},\mathbf{x}) \sim G\left(\begin{bmatrix}\mathbf{m_S}\\ \mathbf{m_X}\end{bmatrix}, \begin{bmatrix}\mathbf{V_S} & \mathbf{V_{SX}}\\ \mathbf{V_{SX}}^T & \mathbf{V_X}\end{bmatrix}\right) \tag{1.88}$$

being $\mathbf{m_S}$ and $\mathbf{m_X}$ the means of $\mathbf{S}$ and $\mathbf{X}$, respectively, $\mathbf{V_S}$ and $\mathbf{V_X}$ the covariance matrix of $\mathbf{S}$ and $\mathbf{X}$, respectively, and $\mathbf{V_{SX}}$ the matrix of crossed covariances of $\mathbf{S}$ and $\mathbf{X}$, that is,

$$\mathbf{V_S} = \mathbb{E}\{(\mathbf{S}-\mathbf{m_S})(\mathbf{S}-\mathbf{m_S})^\intercal\} \tag{1.89}$$
$$\mathbf{V_X} = \mathbb{E}\{(\mathbf{X}-\mathbf{m_X})(\mathbf{X}-\mathbf{m_X})^\intercal\} \tag{1.90}$$
$$\mathbf{V_{SX}} = \mathbb{E}\{(\mathbf{S}-\mathbf{m_S})(\mathbf{X}-\mathbf{m_X})^\intercal\} \tag{1.91}$$

The calculation of the posterior distribution of $\mathbf{S}$ given $\mathbf{X}$ is more complex than in the one-dimensional case, but it follows a similar procedure, which we will omit here. It can be shown that the posterior distribution is Gaussian with mean

$$\mathbf{m_{S|X}} = \mathbf{m_S} + \mathbf{V_{SX}}\mathbf{V_X}^{-1}(\mathbf{x}-\mathbf{m_X}) \tag{1.92}$$

and covariance

$$\mathbf{V_{S|X}} = \mathbf{V_S} - \mathbf{V_{SX}}\mathbf{V_X}^{-1}\mathbf{V_{SX}}^T \tag{1.93}$$

Since the minimum MSE estimator of $\mathbf{S}$ given $\mathbf{X}$ is precisely the posterior mean, we can write

$$\hat{\mathbf{s}}_{\text{MSE}} = \hat{\mathbf{s}}_{\text{MAD}} = \hat{\mathbf{s}}_{\text{MAP}} = \mathbf{m_{S|X}} = \mathbf{m_S} + \mathbf{V_{SX}}\mathbf{V_X}^{-1}(\mathbf{x}-\mathbf{m_X}) \tag{1.94}$$

1.5.3 Linear estimation and Gaussian estimation

Note that, if the target variable is scalar, (1.94) is identical to (1.73). This is not coincidental: if the minimum MSE estimator in the Gaussian case is linear, it must be equal to the best linear estimator, which is given by (1.94).

1.6 ML estimation of probability distributions parameters

Sometimes we may be interested in estimating the parameters of a probability distribution, such as the mean or variance of a Gaussian distribution, the decay parameter that characterizes an exponential distribution, or values a and b delimiting the interval in which a uniform distribution is defined.

In these cases, the prior distribution of these variables is not usually known, even more, in many cases, these parameters are said to be deterministic and are not treated as random parameters. However, if a set of observations generated from these distributions is available, we can obtain the likelihood of these variables and estimate their values with maximum likelihood criteria.

Example 1.10 (ML estimate of the mean and variance of a one-dimensional Gaussian distribution)
The weight of individuals in a family of molluscs is known to obey a Gaussian distribution, but the mean and variance are unknown. The weight of l individuals taken independently, $\{X^{(k)}\}_{k=1}^{l}$, is available.

The likelihood of the mean and the variance for a single observation x, in this case, is given by:

$$p_X(x) = p_{X|m,v}(x|m,v) = \frac{1}{\sqrt{2\pi v}} \exp\left[-\frac{(x-m)^2}{2v}\right] \tag{1.95}$$

Since we must construct the estimator based on the joint observation of l observations, we will need to calculate the joint distribution of all of them which, being independent observations, is obtained as the product of individual observations:

$$\begin{aligned} p_{\{X^{(k)}\}|m,v}(\{x^{(k)}\}|m,v) &= \prod_{k=1}^{l} p_{X|m,v}(x^{(k)}|m,v) \\ &= \frac{1}{(2\pi v)^{l/2}} \prod_{k=1}^{l} \exp\left[-\frac{(x^{(k)}-m)^2}{2v}\right] \end{aligned} \tag{1.96}$$

The ML estimators of m and v will be the values maximizing (1.96). The analytical form of this function suggests the use of the logarithm to simplify the maximization:

$$L = \ln\left[p_{\{X^{(k)}\}|m,v}(\{x^{(k)}\}|m,v)\right] = -\frac{l}{2}\ln(2\pi v) - \frac{1}{2v}\sum_{k=1}^{l}(x^{(k)}-m)^2 \tag{1.97}$$

Differentiating (1.97) with respect to m and v, we get the system of equations to solve

$$\begin{aligned} \left.\frac{d\,L}{d\,m}\right|_{\substack{m=\hat{m}_{\mathrm{ML}} \\ v=\hat{v}_{\mathrm{ML}}}} &= -\frac{1}{v}\sum_{k=1}^{l}(x^{(k)}-m)\Bigg|_{\substack{m=\hat{m}_{\mathrm{ML}} \\ v=\hat{v}_{\mathrm{ML}}}} = 0 \\ \left.\frac{d\,L}{d\,v}\right|_{\substack{m=\hat{m}_{\mathrm{ML}} \\ v=\hat{v}_{\mathrm{ML}}}} &= -\frac{l}{2v} + \frac{1}{2v^2}\sum_{k=1}^{l}(x^{(k)}-m)^2\Bigg|_{\substack{m=\hat{m}_{\mathrm{ML}} \\ v=\hat{v}_{\mathrm{ML}}}} = 0 \end{aligned} \tag{1.98}$$

Solving for m the first equation we get

$$\hat{m}_{\mathrm{ML}} = \frac{1}{l}\sum_{k=1}^{l} x^{(k)} \tag{1.99}$$

which is the sample average of the observations. On the other hand, we can solve the second equation for v to get

$$\hat{v}_{\mathrm{ML}} = \frac{1}{l}\sum_{k=1}^{l} (x^{(k)} - \hat{m}_{\mathrm{ML}})^2 \tag{1.100}$$

which is the sample variance. Note that, if instead of applying the estimation function (of m or v) on some specific observations we did it on generic values $\{X^{(k)}\}$, the estimators could be treated as random variables, i.e.,

$$\hat{M}_{\mathrm{ML}} = \frac{1}{l}\sum_{k=1}^{l} X^{(k)} \tag{1.101}$$

$$\hat{V}_{\mathrm{ML}} = \frac{1}{l}\sum_{k=1}^{l} [X^{(k)} - \hat{M}_{\mathrm{ML}}]^2 \tag{1.102}$$

1.7 Some key results on Estimation Theory

In estimation theory, an estimator should ideally satisfy certain desirable properties to ensure its reliability and optimality. This section introduces three fundamental concepts: *consistency*, the *Cramér-Rao lower bound*, and *efficiency*.

Throughout this section, we assume that the parameter S is **deterministic** (frequentist setting), meaning it is an unknown but fixed quantity that does not vary across different samples.

1.7.1 Consistency

An estimator is consistent if it converges to the true parameter as the sample size increases. This property ensures that, with enough data, the estimator provides an accurate estimate.

Definition 1.3 (Mean Square Consistency) An estimator $\hat{S}_n$ of a target variable S, based on a set on n i.i.d. observational variables $X_0, \ldots, X_{n-1}$ is **mean square consistent** (or, simply, consistent) if

$$\lim_{n\to\infty} \mathbb{E}\{(\hat{S}_n - S)^2 | S = s\} = 0. \tag{1.103}$$

Using the bias-variance decomposition in (1.13), we can conclude that mean square consistency is equivalent to require the variance to vanish as $n \to \infty$, and the estimator to be asymptotically unbiased, that is:

$$\lim_{n\to\infty} \text{var}\{\hat{S}_n|S=s\} = 0, \tag{1.104}$$

$$\lim_{n\to\infty} \text{bias}\{\hat{S}_n|S=s\} = 0. \tag{1.105}$$

1.7.2 Cramér-Rao Lower Bound

The Cramér-Rao lower bound (CRLB) provides a fundamental limit on the variance of any unbiased estimator, establishing the best possible precision an estimator can achieve.

Theorem 1.2 (Cramér-Rao Lower Bound) *Let $\hat{S}$ be an unbiased estimator of a parameter S based on a sample $X_0, X_1, \ldots, X_{n-1}$ with likelihood function $p_{X|S}(x|s)$. The conditional variance of any unbiased estimator satisfies:*

$$var\{\hat{S}|S=s\} \geq \frac{1}{I(s)}, \tag{1.106}$$

*where $I(s)$ is the **Fisher information**, given by:*

$$I(s) = \mathbb{E}\left\{\left(\frac{\partial}{\partial s}\log p_{X|S}(x|s)\right)^2\right\}. \tag{1.107}$$

This result establishes a theoretical lower bound on the variance of an unbiased estimator, providing a benchmark for evaluating estimator performance.

1.7.3 Efficiency

An estimator is efficient if it attains the Cramér-Rao lower bound, meaning it achieves the minimum possible variance among all unbiased estimators.

Definition 1.4 (Efficiency) An unbiased estimator $\hat{S}$ of S is said to be **efficient** if it attains the Cramér-Rao lower bound:

$$\text{var}\{\hat{S}|S=s\} = \frac{1}{I(s)}. \tag{1.108}$$

An estimator is **asymptotically efficient** if its variance asymptotically approaches the CRLB, i.e.,

$$\lim_{n\to\infty} n \cdot \text{Var}(\hat{S}_n) = \frac{1}{I(S)}. \tag{1.109}$$

Efficiency is a desirable property since it ensures that an estimator makes optimal use of the available information, achieving the lowest possible variance.

Theorem 1.3 (Asymptotic Efficiency of Maximum Likelihood Estimation) *Let $\hat{S}_{ML}$ be the maximum likelihood estimator of a deterministic parameter S, based on an i.i.d. sample $X_1, X_2, \ldots, X_n$ with likelihood function $p_{X|S}(x|s)$. Under regularity[a] conditions, the maximum likelihood estimate is asymptotically efficient.*

[a] The regularity conditions are assumptions that ensure the MLE behaves well as the sample size increases. This refers to some conditions on the differentiability of the log-likelihood function, and others.

1.8 Problems

1.1 The posterior distribution of S given X is

$$p_{S|X}(s|x) = x^2 \exp(-x^2 s), \qquad s \geq 0$$

Compute estimators $\hat{S}_{\text{MSE}}$, $\hat{S}_{\text{MAD}}$ y $\hat{S}_{\text{MAP}}$.

1.2 Consider an estimation problem given by the following posterior distribution:

$$p_{S|X}(s|x) = x \exp(-xs), \;\; s > 0 \tag{1.110}$$

Compute estimators $\hat{S}_{\text{MSE}}$, $\hat{S}_{\text{MAD}}$ y $\hat{S}_{\text{MAP}}$.

1.3 A r.v. S must be estimated from the observation of another r.v. X by means of a linear mean square error estimator given by:

$$\hat{S}_{\text{LMSE}} = w_0 + w_1 X$$

Knowing that $\mathbb{E}\{X\} = 1$, $\mathbb{E}\{S\} = 0$, $\mathbb{E}\{X^2\} = 2$, $\mathbb{E}\{S^2\} = 1$ y $\mathbb{E}\{SX\} = 1/2$, compute:

a) The values for w_0 y w_1.

b) The mean square error of the estimator, $\mathbb{E}\left\{\left(S - \hat{S}_{\text{LMSE}}\right)^2\right\}$.

1.4 (Linear estimation of minimum mean squared error) We want to construct a linear estimator of minimum mean squared error that will allow us to estimate the random variable S from the random variables X_1 and X_2. Knowing that

$$\begin{array}{lll} \mathbb{E}\{S\} = 1/2 & \mathbb{E}\{X_1\} = 1 & \mathbb{E}\{X_2\} = 0 \\ \mathbb{E}\{S^2\} = 4 & \mathbb{E}\{X_1^2\} = 3/2 & \mathbb{E}\{X_2^2\} = 2 \\ \mathbb{E}\{SX_1\} = 1 & \mathbb{E}\{SX_2\} = 2 & \mathbb{E}\{X_1 X_2\} = 1/2 \end{array}$$

get the weights from the desired estimator and calculate its squared mean error. Calculate the estimated value for the observation vector: $[X_1, X_2] = [3, 1]$.

1.5 Let X and S be two random variables with joint pdf

$$p_{X,S}(x,s) \begin{cases} 2 \; 0 < x < 1, 0 < s < x \\ 0 \text{ resto} \end{cases}$$

a) Compute the minimum mean square error estimate of S given X, $\hat{S}_{\text{MSE}}$.
b) Compute the risk of estimator $\hat{S}_{\text{MSE}}$.

1.6 A digitized image of dimensions $8x8$ is available, whose luminance values are statistically independent and evenly distributed between 0 (white) and 1 (black); the image has been modified by applying a transformation of the form $Y = X^r$ on each pixel; $r > 0$, where X is the r.v. associated with the pixels of the original image and Y is associated with the transformed image. Obtain the expression that allows estimating r by maximum likelihood given the 64 pixel values of the transformed image $\{y^{(k)}\}_{k=1}^{64}$, without knowing the original image.

1.7 For the design of a communication system, it is desired to estimate the signal attenuation between the transmitter and the receiver, as well as the noise power introduced by the channel when this noise is Gaussian of zero mean and independent of the transmitted signal. For this, the transmitter sends a signal with a constant amplitude of 1 and the receiver collects a set of K observations available at its input.

a) Estimate the channel attenuation, α, and the noise variance, v_r, by maximum likelihood, when the available observations on the receiver are

$$\{0.55, 0.68, 0.27, 0.58, 0.53, 0.37, 0.45, 0.53, 0.86, 0.78\}.$$

b) If the system is to be used for the transmission of digital signals with unipolar coding (a A signal level is used to transmit a bit 1 and the signal level is maintained at 0 for the transmission of bit 0), considering equiprobability between symbols, indicate the minimum level of signal that should be used in the coding, $A_{\min}$, to guarantee a SNR level in the receiver of 3 dB.

1.8 Company *Like2Call* offers hosting services for call centers. In order to dimension the staff of operators the company is designing a statistical model to characterize the activity in the hosted call centers. One of the components of such a model relies on the well-known fact that the times between incoming calls follow an exponential distribution

$$p_{X|S}(x|s) = s\ \exp(-s\,x), \qquad x > 0$$

where random variable X represents the time before a new call arrives, and S is a parameter that depends on the time of the day and each particular call-center service (e.g., attention to the clients of an insurance company, customers of an online bank, etc).

For random variable S, the following *a priori* model is assumed:

$$p_S(s) = \exp(-s), \qquad s > 0.$$

With this information, we would like to design an estimator of S that is based on the first K incoming calls for each implemented service and time interval, i.e., represented by a vector of i.i.d. observations $\mathbf{X} = [X_0, X_1, \cdots, X_{K-1}]$,.

a) Obtain the maximum likelihood estimator or S based on the observation vector $\mathbf{X}$ and verify that it depends just on the sum of all observations, $z = \sum_{k=0}^{K-1} x_k$.
b) Compute the posterior distribution of S given $\mathbf{X}$, $p_{S|\mathbf{X}}(s|\mathbf{x})$.

c) Obtain the maximum *a posteriori* estimator of S given $\mathbf{X}$, $\hat{s}_{\text{MAP}}$.
d) Obtain the minimum mean square error estimator of S given $\mathbf{X}$, $\hat{s}_{\text{MSE}}$.
e) Calculate the mean square error given $\mathbf{X}$ of a generic estimator $\hat{S}$, and particularize the result for estimators of the following analytical shape $\hat{s}_c = \frac{c}{z+1}$.
f) Find expressions for $p_{Z|S}(z|s)$, $p_{Z,S}(z,s)$, and $p_Z(z)$.
g) Calculate the mean square error of a generic estimator $\hat{s}_c = \frac{c}{z+1}$. Study how the result changes with c and K.

You can use the following results:

i.
$$\int_0^\infty x^N \exp(-x)dx = N!$$

ii. If $f(x) = a\, exp(-a\, x)$, $x > 0$ then

$$\underbrace{f(x) * f(x) * \cdots * f(x)}_{N \text{ times}} = \frac{a^N\, x^{N-1}}{(N-1)!} exp(-a\, x),\ x > 0$$

iii. For K an integer
$$\int_0^\infty \frac{K\, x^{K-1}}{(x+1)^{K+3}} dx = \frac{2}{(K+2)(K+1)}$$

Solution 1.1 (Solution to problem 2.8)

a) Since $p_{\mathbf{X}|S}(\mathbf{x}|s) = s^K\ \exp(-s\, z)$, with $z > 0$, we have $\ln p_{\mathbf{X}|S}(\mathbf{x}|s) = K \ln s - s\, z$ and, thus,

$$\frac{d}{ds} \ln p_{\mathbf{X}|S}(\mathbf{x}|s) = \frac{K}{s} - z$$

Therefore,

$$\hat{s}_{\text{ML}} = \frac{K}{z}$$

b) The joint distribution of S and X is

$$p_{\mathbf{X},S}(\mathbf{x},s) = p_{\mathbf{X}|S}(\mathbf{x}|s) p_S(s) = s^K\ \exp[-s(z+1)]$$

therefore

$$p_{\mathbf{X}}(\mathbf{x}) = \int p_{\mathbf{X},S}(\mathbf{x},s)\, ds = \int_0^\infty s^K\ \exp[-s(z+1)]\, ds$$

With the change of variable $s' = s(z+1)$ the integral can be simplified using expression (i), and we get

$$p_{S|\mathbf{X}}(s|\mathbf{x}) = \frac{p_{\mathbf{X},S}(\mathbf{x},s)}{p_{\mathbf{X}}(\mathbf{x})} = \frac{(z+1)^{K+1}\ p_{\mathbf{X},S}(\mathbf{x},s)}{K!} = \frac{s^K (z+1)^{K+1}\ \exp[-s(z+1)]}{K!}$$

c) The MAP estimate is

$$\begin{aligned}
&\hat{s}_{\text{MAP}} = \arg\max_s p_{S|\mathbf{X}}(s|\mathbf{x}) = \arg\max_s p_{\mathbf{X},S}(\mathbf{x},s) \\
&\ln p_{\mathbf{X},S}(\mathbf{x},s) = K \ln s - s\,(z+1) \\
&\frac{d}{ds} \ln p_{\mathbf{X},S}(\mathbf{x},s) = \frac{K}{s} - (z+1) \\
&\hat{s}_{\text{MAP}} = \frac{K}{z+1}
\end{aligned}$$

d) The minimum MSE estimate is

$$\hat{s}_{\text{MSE}} = \mathbb{E}\{S|\mathbf{x}\} = \int s\, p_{S|\mathbf{X}}(s|\mathbf{x})\, ds = \frac{(z+1)^{K+1}}{K!} \int_0^\infty s^{K+1}\, \exp[-s(z+1)]\, ds$$

Replacing again $s' = s(z+1)$ and using expression (i), we get

$$\hat{s}_{\text{MSE}} = \frac{K+1}{z+1}$$

e) The calculation is somehow tedious, but can be summarized as follows:

$$\begin{aligned}
\mathbb{E}\left\{(S-\hat{s})^2|X\right\} &= \int_0^\infty (s-\hat{s})^2\, p_{S|X}(s|x) ds \\
&= \frac{(z+1)^{K+1}}{K!}\left[\frac{(K+2)!}{(z+1)^{K+3}} + \hat{s}^2 \frac{K!}{(z+1)^{K+1}} - 2\hat{s}\frac{(K+1)!}{(z+1)^{K+2}}\right] \\
&= \frac{(K+2)(K+1)+c^2-2c(K+1)}{(z+1)^2}
\end{aligned}$$

For the MAP and MSE estimators, the expressions are substantially simplified:

$$\begin{aligned}
\mathbb{E}\left\{(S-\hat{s}_{MAP})^2|z\right\} &= \frac{K+2}{(z+1)^2} \\
E\left\{(S-\hat{s}_{MSE})^2|z\right\} &= \frac{K+1}{(z+1)^2}
\end{aligned}$$

f) Using the fact that Z is the sum of K i.i.d. variables (given S):

$$p_{Z|S}(z|s) = \underbrace{[s\, \exp(-s\, z)] * \cdots * [s\, \exp(-s\, z)]}_{K \text{ times}} = \frac{s^K\, z^{K-1}}{(K-1)!} \exp(-s\, z), \qquad z > 0$$

The joint pdf of Z and S can now be obtained as

$$p_{Z,S}(z,s) = p_{Z|S}(z|s) p_S(s) = \frac{s^K\, z^{K-1}}{(K-1)!} \exp[-s\,(z+1)], \qquad s,z > 0$$

Finally, integrating s out, we have

$$p_Z(z) = \int p_{Z,S}(z,s) ds = \frac{z^{K-1}}{(K-1)!} \int_0^\infty s^K \exp[-s\,(z+1)] = \frac{K\, z^{K-1}}{(z+1)^{K+1}},\ \ z > 0$$

g)

$$\mathbb{E}\left\{(S-\hat{S}_c)^2\right\} = \int \mathbb{E}\left\{(S-\hat{s}_c)^2|z\right\}\, p_Z(z) dz$$

Using the results from the previous two sections we can obtain

$$\mathbb{E}\left\{(S-\hat{S}_c)^2\right\} = \left[(K+2)(K+1)+c^2-2c(K+1)\right] \int_0^\infty \frac{K\, z^{K-1}}{(z+1)^{K+3}} dz$$

The value of the integral is given in (iii). Simplifying also for the MAP and MSE estimators:

$$\begin{aligned}
\mathbb{E}\left\{(S-\hat{S}_{MAP})^2\right\} &= \frac{2}{K+1} \\
E\left\{(S-\hat{S}_{MSE})^2\right\} &= \frac{2}{K+2}
\end{aligned}$$